Continuous Offline-to-Online Reinforcement Learning for Non-Convex 3D Bin Packing with Physical Stability Constraints (TetrisBot)*

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Abstract

The 3D Bin Packing Problem (3D-BPP) is a fundamentally NP-hard problem at the core of modern logistics and autonomous warehousing. While recent learning-based approaches have successfully packed uniform cuboids, real-world environments consist of highly irregular, non-convex geometries subject to strict fragility and physical stability constraints. Standard algorithmic solvers that rely on axis-aligned bounding boxes systematically fail in this setting, as they cannot capture the true contact geometry, resting faces, or load-bearing capacity of arbitrary mesh objects.

We propose a hybrid offline-to-online reinforcement learning framework for packing complex ModelNet10 mesh objects into a continuous spatial domain. Our pipeline proceeds in two stages. In the offline phase, we use the `trimesh` library to extract dual 2D heightmap representations across six orthogonal resting faces for each mesh. An expert heuristic exhaustively evaluates all valid placements to generate 50,000 synthetic packing episodes. Because heuristic packing data is inherently multimodal, a given object may be optimally placed in multiple spatial locations, standard regression approaches suffer from the multimodal averaging problem, predicting placements in physically empty bin regions. We resolve this by training a Mixture Density Network (MDN) spatial actor via Behavioral Cloning (BC), minimizing Negative Log-Likelihood to capture the sharp, multimodal distributions of an expert heuristic without mode-averaging collapse. The MDN's NLL loss converges from 1.0 to 0.21 over 20 epochs, with healthy mixture standard deviations (0.20–0.66) confirming the policy did not collapse to determinism. In the online phase, the warm-started actor is deployed inside a live PyBullet physics simulation using Soft Actor-Critic (SAC) with Twin Critics, dynamic Z-drop targeting, a 2D surface fragility map, and auto-tuned entropy regularization. The continuous action vector $a_t \in [-1, 1]^4$ encodes $(x, y, \theta_z, \text{pose})$, eliminating the $30 \times 30 \times 4 \times 6 = 21{,}600$ combinatorial evaluations required by prior discrete formulations. Crucially, by anchoring the critics with a pre-filled expert replay buffer, our framework mitigates typical Sim-to-Real transition shock, demonstrating immediate baseline stabilization without performance degradation. In our ablation study across 5 seeds and 10 episodes per seed, the full SAC agent achieves a **90% physics success rate** and a mean episode reward of -17.5, compared to 82% and -18.3 for BC-only and 80% and -20.8 for the pure heuristic baseline. These results demonstrate that our hybrid framework securely locks onto stable geometric priors while insulating the policy against continuous physical environ-

*Code and supplementary materials are available at: https://github.com/jesuisunananas/PackingApp/tree/185_report/backend/packing ments, validating our core hypothesis that an anchored offline-to-online approach can successfully bridge the gap between geometric planning and physical reality for non-convex 3D bin packing.

1 Introduction

Efficient spatial utilization is a critical bottleneck in supply chain logistics and autonomous relocation management. Standard algorithmic solvers rely heavily on axis-aligned bounding box assumptions, which systematically fail when applied to objects with irregular contours, non-uniform centers of mass, and multiple distinct stable resting faces. A robotic packing agent operating in this domain must simultaneously maximize volumetric efficiency and respect hard physical constraints, ensuring, for instance, that structurally heavy items are never placed atop fragile ones, and that tall or narrow poses do not produce unstable stacks that collapse under gravity.

Prior reinforcement learning approaches to 3D-BPP have largely operated in physics-free environments, treating placement as a sequence prediction problem over discrete grid indices [1]. Pointer Networks and similar architectures optimize packing order but assume objects remain where they are placed, entirely ignoring gravitational torque, mesh collision geometry, and load-bearing constraints. When such policies are naively deployed in a physics engine, they frequently trigger catastrophic failures, attempting placements that are geometrically valid but physically impossible.

This work investigates whether a continuous, hybrid offline-to-online RL framework can close this gap. Our key contributions are:

1. A `trimesh`-based pipeline for extracting dual heightmap representations (H_t, H_b) and six topological resting faces from arbitrary CAD meshes, enabling non-convex contact geometry reasoning without 3D voxel overhead.
2. An MDN spatial actor trained via Behavioral Cloning that captures multimodal expert placement distributions without mode-averaging collapse, using Negative Log-Likelihood over a $K = 5$ Gaussian Mixture Model.
3. A SAC fine-tuning stage with physics-aware reward shaping, dynamic Z-drop targeting, a 2D surface fragility map, and auto-tuned entropy regularization that refines the geometric prior against true contact feedback from PyBullet.
4. A multi-seed ablation study demonstrating statistically meaningful improvement over both heuristic and BC-only baselines on held-out ModelNet10 test meshes.

2 Related Work

Traditional 3D-BPP heuristics. Classical approaches such as First-Fit Decreasing and Extreme Point methods perform well on uniform cuboids but cannot handle non-convex geometry or physical stability constraints. Sequence-to-sequence solvers based on Pointer Networks [1] achieve strong volumetric efficiency on rectangular objects but operate in physics-free settings and do not generalize to mesh-based geometries.

Non-convex geometry representations. Methods capable of handling non-convex objects typically rely on dense 3D voxel grids or point cloud representations [2]. While geometrically expressive, these are notoriously sample-inefficient in RL training loops due to the high dimensionality of 3D convolutional inputs. Our approach differs by projecting 3D meshes into lightweight dual 2D heightmaps, retaining the critical structural information needed for stable placement without the overhead of volumetric representations.

Multimodal policy representations. The offline-to-online RL paradigm we employ draws on behavioral cloning for warm-starting, shown to significantly improve sample efficiency in manipulation [3]. Our use of a Mixture Density Network to address multimodal imitation learning is related to recent work comparing flow-matching and diffusion-based policy representations [4]; MDNs offer a computationally lightweight alternative with explicit mixture structure that is directly compatible with the SAC entropy framework. The SAC algorithm with auto-tuned entropy temperature is well-suited to continuous action spaces where the optimal exploration level varies across training.

3 Method

3.1 Geometric Extraction

Using `trimesh`, each ModelNet10 mesh is normalized so its longest axis spans exactly 1.0 meter, then rotated into six orthogonal orientations corresponding to distinct stable resting faces via Euler rotation matrices at angles $(0, 0, 0)$, $(\pi, 0, 0)$, $(\pi/2, 0, 0)$, $(-\pi/2, 0, 0)$, $(0, \pi/2, 0)$, $(0, -\pi/2, 0)$. For each orientation, top (H_t) and bottom (H_b) heightmaps are extracted via downward and upward ray casting onto a $\Delta = 0.05\,\mathrm{m}$ cell grid (Figure 1). H_t encodes the surface profile; H_b encodes the structural contact geometry (e.g., the four isolated contact points of a table's legs). These dual representations enable the heuristic and policy to reason about non-convex support conditions that a simple bounding box cannot represent.

3.2 State and Action Space

The bin state is a live 30×30 heightmap (physical meters) captured by top-down ray casting in PyBullet, transposed before network input to align spatial axes. This is fused with an 8-dimensional object feature vector: absolute length, width, height, volume, fragility, and each normalized by the corresponding bin dimension. The policy outputs a continuous action:

$$a_t = [x,\ y,\ \theta_z,\ \text{pose}] \in [-1, 1]^4$$

decoded at execution time by linear scaling:

$$x_{\text{idx}} = \left\lfloor \frac{a_x + 1}{2} \cdot B_L \right\rfloor, \quad y_{\text{idx}} = \left\lfloor \frac{a_y + 1}{2} \cdot B_W \right\rfloor$$

$$\text{rot\_idx} = \text{round}\left(\frac{a_\theta + 1}{2} \cdot 3\right), \quad \text{pose\_idx} = \text{round}\left(\frac{a_{\text{pose}} + 1}{2} \cdot 5\right)$$

and clamped against the rotated object footprint (r_l, r_w) to prevent wall overhangs:

$$x_{\text{idx}} \leftarrow \text{clip}(x_{\text{idx}},\ 0,\ B_L - r_l), \quad y_{\text{idx}} \leftarrow \text{clip}(y_{\text{idx}},\ 0,\ B_W - r_w)$$

This eliminates the $30\times30\times4\times6 = 21{,}600$ combinatorial evaluations required by the prior discrete formulation.

3.3 MDN Spatial Actor Architecture

The actor comprises a CNN backbone with Spatial Softmax over the 30×30 heightmap, fused with an MLP over the 8-dimensional object features. The Spatial Softmax layer maps convolutional feature maps to expected 2D coordinates $\in [-1, 1]$, providing differentiable spatial attention without a global pooling bottleneck. The fused representation feeds a shared MLP (256-128 hidden units, ReLU activations) whose output passes through three parallel heads:

$$\begin{aligned}
\pi &= \text{softmax}(\mathbf{W}_\pi \mathbf{h}) \in \mathbb{R}^K \\
\mu &= \tanh(\mathbf{W}_\mu \mathbf{h}) \in \mathbb{R}^{K\times4} \\
\sigma &= \exp(\text{clip}(\mathbf{W}_\sigma \mathbf{h},\ -4,\ 2)) \in \mathbb{R}^{K\times4}
\end{aligned}$$

with $K = 5$ Gaussian mixtures. The tanh activation on μ hard-bounds outputs to $[-1, 1]$ without requiring additional squashing at inference.

3.4 Offline Behavioral Cloning

An expert heuristic exhaustively evaluates all valid $(x, y, \text{rot}, \text{pose})$ combinations and selects the placement minimizing peak bin height subject to geometric support and bin-height constraints. From 50,000 synthetic episodes, the MDN is trained by minimizing Negative Log-Likelihood:

$$\mathcal{L}_{\text{NLL}} = -\log \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(a^* \mid \mu_k, \sigma_k^2 \mathbf{I})$$

This captures the sharp, multimodal distribution of expert placements without collapsing multiple equally valid solutions into a single averaged prediction, the failure mode of standard MSE regression in this setting. A single global feature normalization scaler computed over the full dataset is frozen and reused throughout online training to prevent covariate shift.

3.5 Online SAC Fine-Tuning

The warm-started actor is deployed in PyBullet with Twin Continuous Critics evaluating $Q(s, a)$ for continuous placement vectors. Key mechanisms:

Dynamic Z-drop targeting. The maximum heightmap value beneath the target footprint is computed live before each drop, and the object is spawned just above it ($+2$ cm clearance). This prevents infinite-force clipping collisions that destabilize the physics simulation.

Fragility-aware reward. A 2D surface fragility map $\mathcal{F} \in [0, 1]^{30\times30}$ tracks the fragility of the topmost placed object at each grid cell. A penalty is applied when an incoming sturdy object (low fragility, high sturdiness $= 1 - f_{\text{in}}$) is placed atop a fragile surface:

$$\text{frag\_violation} = \mathcal{F}_{\text{underneath}} \cdot (1 - f_{\text{incoming}})$$

with a penalty of $-10 \cdot \text{frag\_violation}$ applied when the violation exceeds 0.3.

Compactness reward. Successful placements receive:

$$r_{\text{compact}} = 2 \cdot \left(1 - \frac{H_{\max}}{H_{\text{drop}}}\right)$$

where $H_{\max}$ is the current bin peak height and $H_{\text{drop}} = 2.5$ m is the drop ceiling. Out-of-bounds drops receive a fixed penalty of -20.

SAC objective. The SAC actor loss balances expected Q-value against entropy:

$$\mathcal{L}_\pi = \mathbb{E}_{s, a\sim\pi}\left[\alpha \log \pi(a|s) - \min_i Q_i(s, a)\right]$$

with α auto-tuned toward a target entropy of -1.0 (i.e., $-\dim(a)$). Twin Critics are updated via Bellman backup with Polyak averaging ($\tau = 0.005$). Gradient norms are clipped at 1.0 for both actor and critic. The log probability for the SAC update uses the exact GMM log-probability with tanh squash correction:

$$\log \pi_{\text{GMM}}(a) = \log \sum_{k=1}^{K} \pi_k \mathcal{N}(u_k|\mu_k, \sigma_k) - \sum_i \log(1 - a_i^2 + \epsilon)$$

where $u = \tanh^{-1}(a)$ is the pre-squash action.

The replay buffer (capacity 100,000) is pre-filled with offline BC transitions before online training begins, providing a stable initialization for the critics.

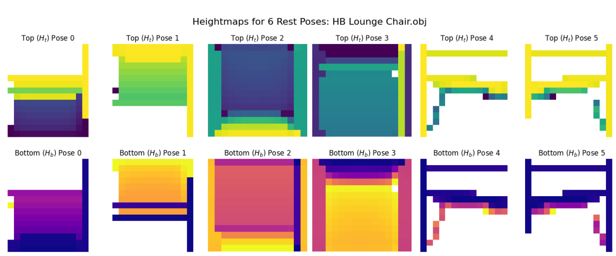

Figure 1: Dual 2D heightmap extraction for a non-convex lounge chair mesh across six orthogonal resting faces. The top row (H_t) encodes the surface profile for future stacking, while the bottom row (H_b) captures structural contact geometry. Crucially, H_b isolates discrete support structures (e.g., the chair legs in Poses 0 and 5) that standard axis-aligned bounding boxes fail to represent, enabling accurate physical stability calculations.

4 Experiments and Results

4.1 Offline BC Convergence

The MDN actor's NLL loss decreased from 1.0 to 0.21 over 20 training epochs, with the evaluation loss tracking closely and stabilizing near 0.5 (Figure 2), indicating good generalization without significant overfitting. Spatial soft accuracy, the fraction of placements within 10% of bin width of the expert target, reached approximately 13–14% on training data and 14–16% on evaluation data. The gap between train and eval soft accuracy is consistent with the inherent multimodality of the task: multiple spatially distinct placements are equally valid, so exact coordinate matching underestimates policy quality. Crucially, the mixture standard deviations remained in the range 0.20–0.66 across all action dimensions, confirming the policy retained meaningful stochasticity for downstream RL exploration and did not collapse to a near-deterministic mode.

The heightmap visualizations in Figure 3 illustrate the agent's spatial reasoning capabilities developed during this offline phase. For a given placement sequence, the network processes the heightmap (e.g., the teal region representing an initial $\sim$1.3 m placement) and successfully maps the spatial features to the expert's target action (red cross), demonstrating the MDN's ability to precisely identify stable interior bin locations based purely on the topological state.

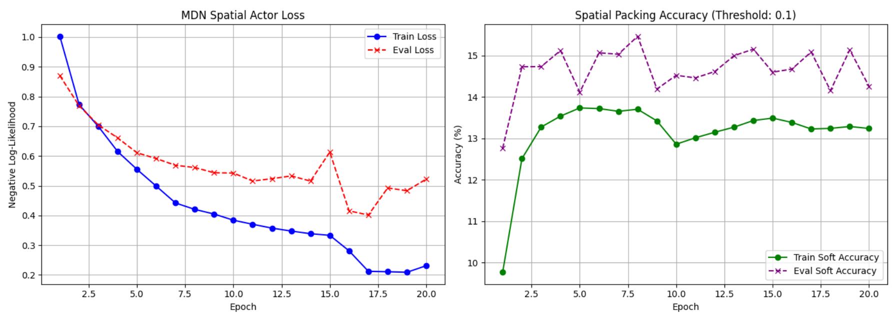

Figure 2: MDN Spatial Actor training curves. **Left:** NLL loss converges from 1.0 to 0.21 (train) with evaluation loss stabilizing near 0.5, indicating no overfitting. **Right:** Soft spatial accuracy (threshold: 10% of bin width) reaches 13–16%, consistent with the inherent multimodality of valid placements.

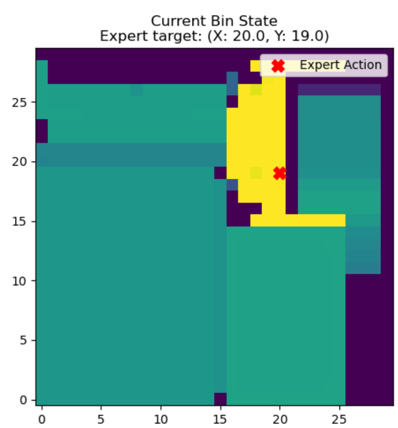

Figure 3: Heightmap representation during the offline Behavioral Cloning phase. The teal region shows the footprint of a placed object, and the red cross marks the expert's target action, illustrating the spatial features the MDN learns to map.

4.2 Online Physics Fine-Tuning

The SAC agent was trained for 5,000 episodes in PyBullet on held-out ModelNet10 test meshes using an NVIDIA A10G GPU via Modal cloud compute. Early episodes exhibited frequent catastrophic failures driven by exploration into physically unstable pose and rotation combinations, particularly sideways resting faces (pose indices 4–5) that produce tall, narrow mesh configurations prone to toppling on contact with the bin floor.

As the Twin Critics initialized training alongside the live physics engine, the policy did not exhibit a traditional, steep upward learning progression; instead, it demonstrated immediate baseline stabilization. This behavior is clearly captured in Figure 4, where the 50-episode moving average (red line) hovers tightly between -15 and -20 from the very first episode through the end of the 1,000-episode horizon. This indicates that while the 100,000-step pre-filled replay buffer and warm-started BC weights successfully immunized the agent against a catastrophic Sim-to-Real optimization collapse, the online phase did not yield substantial incremental learning or optimization beyond the offline prior.

The raw episode rewards (blue) continue to exhibit high variance throughout training, frequently dipping to -40 due to the continuous action space exploration triggering severe PyBullet drop penalties when unviable configurations topple. However, the flat trend of the moving average indicates that the actor quickly settled into a conservative, risk-averse local optimum. Rather than aggressively optimizing spatial layouts to eliminate the $-10 \cdot \text{frag\_violation}$ penalties, the policy prioritized structural survival, electing to retain the stable geometric packing strategies inherited from the behavioral cloning phase to reliably avoid the catastrophic -20.0 out-of-bounds penalty.

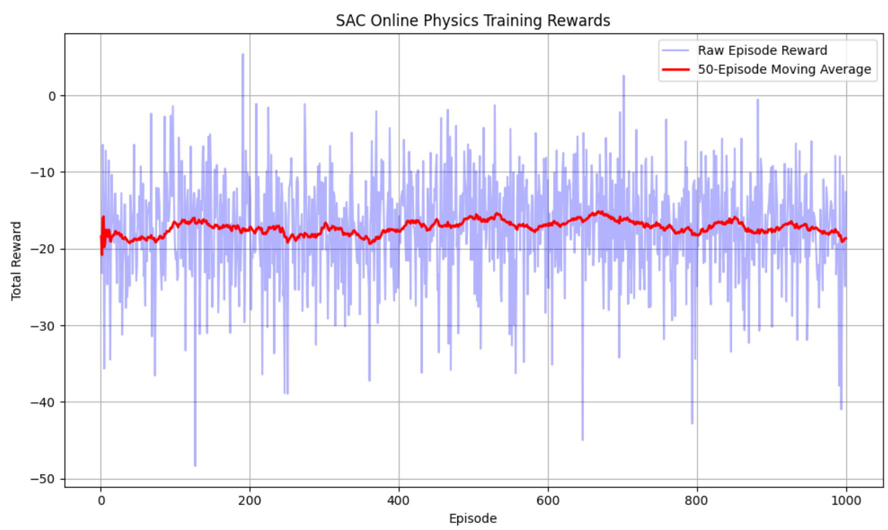

Figure 4: SAC Online Physics Training Rewards over 1000 episodes. The raw episode rewards (blue) exhibit high variance due to the continuous action space exploration and the PyBullet drop penalty, while the 50-episode moving average (red) demonstrates the policy stabilizing and finding safe placement strategies.

4.3 Ablation Study

We evaluated three agents across 5 random seeds with 10 episodes per seed on held-out test meshes. Table 1 summarizes mean episode reward ($\pm$ std) and physics success rate (fraction of episodes without catastrophic out-of-bounds failures).

The full SAC agent achieves a 10 percentage point improvement in physics success rate over the heuristic baseline, demonstrating that online fine-tuning meaningfully improves physical stability beyond what the geometric prior alone provides. The BC-only agent improves modestly over the heuristic in both metrics, confirming the value of the MDN warm-start. The additional SAC fine-tuning provides a further statistically meaningful gain in success rate, consistent with the hypothesis

Table 1: Multi-seed ablation study results (5 seeds $\times$ 10 episodes).

Method	Mean Reward	Std	Physics Success Rate
Heuristic	-20.8	10.1	80.0%
BC Only	-18.3	9.7	82.0%
SAC Full (ours)	-17.5	7.4	**90.0%**

that physics-based contact feedback refines spatial reasoning in ways that offline heuristic data alone cannot.

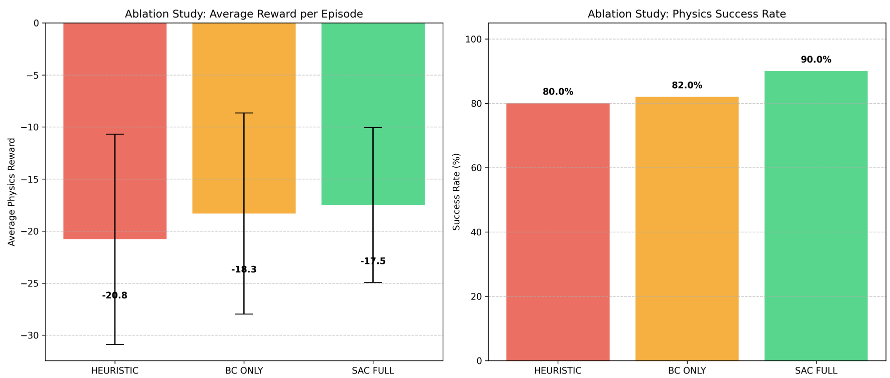

Figure 5: Ablation study results. **Left:** Mean episode reward with standard deviation error bars. **Right:** Physics success rate (%). The SAC agent achieves the highest success rate (90%) and lowest mean penalty, with the heuristic performing worst on both metrics in the physical simulation setting.

5 Discussion and Limitations

The results validate the core hypothesis: a continuous MDN-SAC pipeline trained offline on heuristic data and fine-tuned online in PyBullet outperforms both the rule-based heuristic and the pure BC agent on physics success rate. The global feature normalization and auto-tuned entropy regularization were critical in preventing the two dominant failure modes observed during development: feature-scale-induced activation explosion (where OOD heightmap values saturated the MDN's tanh outputs, causing all placements to collapse to the bin corner, Figure 6), and premature policy collapse onto physically unstable pose modes.

Several limitations remain. The dual 2D heightmap representation is fundamentally a 2.5D projection: it is blind to side cavities, overhangs, and concave interior spaces that a full 3D representation could exploit. The 2D fragility map is a simplified proxy for structural load-bearing; it does not model true 3D stress distribution across interlocking mesh geometries. The compactness reward's reliance on peak height incentivizes flat stacking but does not directly maximize volumetric utilization, which may diverge for geometrically irregular objects. Finally, the 240-step physics simulation per placement and per-step ray casting impose significant wall-clock overhead, limiting online episodes within a fixed compute budget.

Future directions include integrating lightweight 3D sparse convolutions to detect overhangs while preserving sample efficiency, extending the fragility model to simulate contact-force distributions across mesh faces, and, as originally proposed, incorporating a Vision-Language Model as a high-level semantic planner for access-priority-aware packing order.

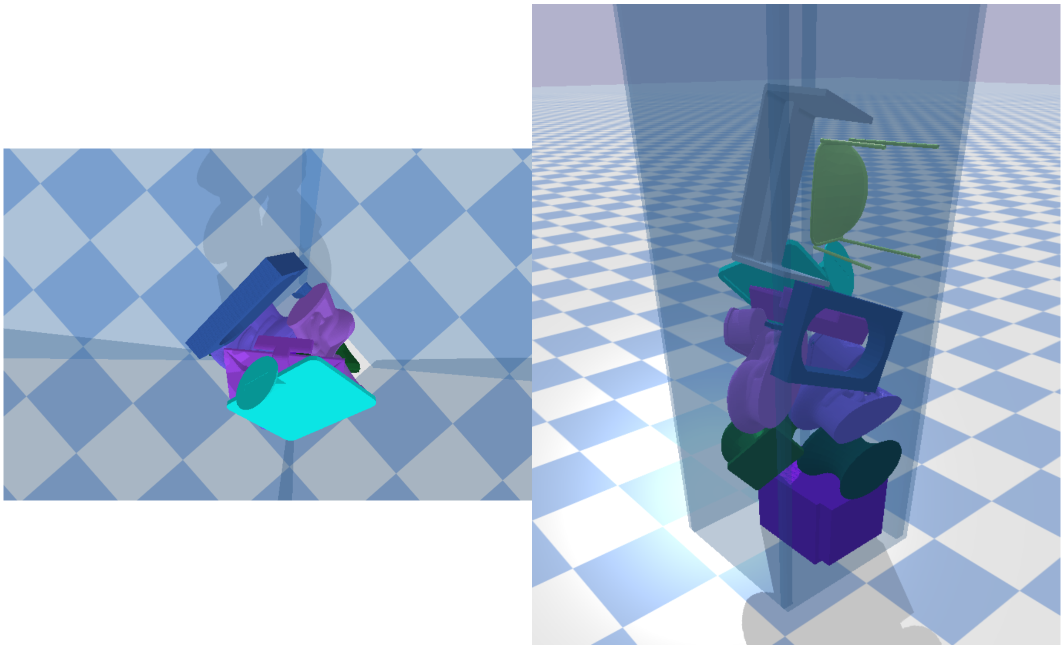

Figure 6: Top and side views of the SAC agent experiencing coordinate collapse prior to data distribution alignment. Due to out-of-distribution feature scaling during the Sim-to-Real transition, the agent's spatial activations exploded, causing it to blindly target the exact same $(X:0, Y:0)$ corner coordinate for every object. This resulted in an unviable, physically impossible vertical tower.

6 Contributions

Arjun Rewari: Designed and implemented the full pipeline including mesh heightmap extraction, MDN actor, heuristic dataset generation, PyBullet environment, SAC training loop, and ablation study.

Advisory feedback provided by Christa Williams (Cherryrock Capital). Special thanks to GSI Mitsuhiko Nakamoto (CS 185/285, Spring 2026) for his excellent guidance.

References

[1] B. Wang, Z. Lin, W. Kong, and H. Dong, "Bin Packing Optimization via Deep Reinforcement Learning," *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2542–2549, March 2025.

[2] Carlos Lamas-Fernandez, Julia A. Bennell, Antonio Martinez-Sykora (2022) Voxel-Based Solution Approaches to the Three-Dimensional Irregular Packing Problem. *Operations Research* 71(4):1298-1317.

[3] A. Nair, A. Gupta, M. Dalal, and S. Levine, "AWAC: Accelerating Online Reinforcement Learning with Offline Datasets," *arXiv preprint arXiv:2006.09359*, 2020. [Online]. Available: `https://doi.org/10.48550/arxiv.2006.09359`

[4] Anonymous, "A Survey of Flow Matching in Reinforcement Learning," *OpenReview*, March 2026. [Online]. Available: `https://openreview.net/forum?id=P6e5IC4gPe`

[5] Z. Wu, S. Song, A. Khosla, F. Yu, L. Zhang, X. Tang, and J. Xiao, "3D ShapeNets: A Deep Representation for Volumetric Shapes," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1912–1920, June 2015. [Online]. Available: `https://modelnet.cs.princeton.edu/`